EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**O(log n) means execution time is literally log n seconds.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

# Search and sort: comparisons and Big O

Name: _____ Class: _____ Date: _____

**1. What property does binary search need that linear search does not?**

**2. What time complexity is typical of merge sort?**

**3. For n=5, how many times does a single for-loop from 0 to n-1 run?**

**4. A nested loop runs 4 times outside and 4 times inside. How many inner operations occur?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Sorted input.

### 2. Model answer

O(n log n).

### 3. Model answer

5 times.

### 4. Model answer

16 operations.

### Vocabulary — know these terms

time complexity — How an algorithm's work grows with input size.

O(n) — Work proportional to input count.

O(log n) — Work proportional to the number of halvings.

O(n²) — Quadratic growth, often nested loops.

trade-off — A benefit balanced against another cost.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Is binary search always better than linear search? Give one counterexample.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Name the complexity of a single loop visiting n values.

### Question 2

An array has 128 ordered items. Compare linear and binary worst-case checks.





NEW LEARNING

# Read, annotate and explain

### Count what grows

Big O describes how work increases with input size, not the exact milliseconds on one computer.

### Compare search

Linear search can take n checks; binary search on sorted input roughly halves the remaining candidates each step.

### Compare sorting

A simple nested comparison loop can be $O(n^2)$, whereas merge sort is $O(n \log n)$.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**For 16 values, compare maximum comparisons of linear and binary search.**

### Predict first

Assume one successful or unsuccessful lookup in sorted data.

### Model solution

Linear search can inspect all 16 elements; binary search checks at most 5 midpoints using this inclusive implementation.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve